

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points. NO CALCULATORS allowed. If you use a calculator, the entire test will receive a score of 0.

Problem 1

Express the area and perimeter of an equilateral triangle as a function of the triangle's side length x .

Problem 2

For each of the following functions, show whether it is even, odd, or neither.

a) $h(t) = 2t + 1$

b) $f(x) = 2x^2 + 1$

Problem 3

CLEARLY graph the function $y = \sqrt{x - 4}$. Actually calculate and plot at least two points.

Problem 4

Let $f(x) = \sqrt{x + 1}$, $g(x) = \frac{1}{x+4}$, and $h(x) = \frac{1}{x}$.

a) Write a formula for $f \circ g \circ h$.

b) Write a formula for $h \circ g \circ f$.

Problem 5

Let $f(x) = 2^3 - x^3$. Give an equation for the graph compressed horizontally by a factor of 3.

Problem 6

On a circle of radius 15 m, how long is an arc that subtends a central angle of (a) $\frac{4\pi}{5}$ radians? (b) 210° ?

Problem 7

In the following, one of $\sin(x)$, $\cos(x)$, and $\tan(x)$ is given. Find the other two if x lies in the specified interval.

a) $\sin(x) = \frac{4}{5}$, $x \in [\frac{\pi}{2}, \pi]$.

b) $\cos(x) = \frac{5}{13}$, $x \in [-\frac{\pi}{2}, 0]$.

Problem 8

Evaluate $\cos(\frac{\pi}{12})$.

Extra Credit:

Prove that if a , b , and c are the sides opposite the angles A , B , and C in a triangle, then

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}.$$